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# A comprehensive review of deep learning models for denoising EEG signals: challenges, advances, and future directions

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**Abstract**
Recent technological advancements have led to a significant increase in electroencephalogram (EEG)-based applications, ranging from clinical diagnosis and brain computer interfaces (BCI) to sleep studies and the monitoring of cognitive tasks. However, raw EEG signals are highly susceptible to artifacts, which hinder accurate analysis and interpretation of brain signals. Traditional artifact removal techniques often fall short owing to their linear assumptions and limited generalizability. Deep learning (DL) based approaches have shown remarkable potential for capturing nonlinear and complex features of EEG signals. This review provides a comprehensive overview of state-of-the-art deep learning models developed for EEG denoising, highlighting their architectural designs, strengths, limitations, and performance analysis. In addition, the study emphasizes the importance of accurate evaluation frameworks and benchmarking, as well as the challenges of generalizability and interpretability. Future research directions include the integration of hybrid architectures, self-supervised learning, and real-time implementation. This article serves as a resource for researchers aiming to advance EEG denoising through modern deep learning approaches.

**Keywords** Artifact removal, Autoencoders, Convolutional neural networks, Electroencephalogram, Emerging trends, Generative adversarial networks, Key challenges, Recurrent neural networks, Transformer models

## 1 Introduction

Electroencephalography (EEG) is a noninvasive technique that captures the brain's electrical activity through electrodes placed on the scalp [1]. It offers a high temporal resolution, enabling the detection of fast changing cognitive and neuronal processes. EEG plays a vital role in neuroscience [2], clinical diagnosis [3], brain computer interface (BCI) applications [4], and biometric identification [5]. It is commonly used to monitor neurological disorders such as epilepsy and sleep disorders [6, 7], and to assess mental health conditions, including depression [8], and Alzheimer's disease [9]. Despite their advantages, EEG signals, characterized by microvolt-range amplitudes, are highly

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susceptible to contamination by various artifacts [10]. These include physiological sources (for instance, eye blinks, muscle activity, cardiac signals) and nonphysiological sources (for instance, electrode pop, power line interference) [11]. Such artifacts often overlap spectrally and temporally with actual brain activity, making their separation and removal challenging. Ocular artifacts can reach amplitudes up to ten times greater than the EEG signal and typically affect frontal electrodes [12, 13]. Muscle activity introduces a wide range of noise (0–200 Hz), distorting high frequency bands such as beta and gamma [14]. Cardiac artifacts, with similar frequencies and amplitudes to EEG, further confuse signal interpretation [15]. Such contaminations degrade the interpretability of EEG data, potentially leading to false alarms in clinical systems or errors in BCI control. Thus, artifact removal (denoising) is a crucial preprocessing step to preserve the quality of neural signals [16–18].

While nonphysiological artifacts can often be reduced by proper electrode placement, shielding, and grounding, physiological artifacts are unavoidable and require algorithmic approaches [19–22]. Traditional signal processing techniques, such as linear regression [23] and adaptive filtering methods [24], require additional reference signals and suffer from mutual contamination issues. Source decomposition methods, such as wavelet transform (WT) and empirical mode decomposition (EMD) techniques, provide a higher frequency resolution but depend on threshold parameters [25]. Blind source separation methods, such as independent component analysis (ICA) [26], canonical correlation analysis (CCA) [27], principal component analysis (PCA) [28], and morphological component analysis (MCA) [29] depends on various statistical assumptions, such as independence, correlation, covariance, and morphology, to extract artifact sources. Such assumptions may not hold true for real EEG data. In addition, traditional methods often struggle to model the nonlinear and dynamic properties of artifacts and neural signals.

To overcome these limitations, researchers have proposed hybrid methods that combine various techniques. These include ICA with wavelet transform [30, 31], adaptive filtering with wavelets [32], and ICA combined with machine learning classifiers [33–35]. In addition, several studies have explored hybrid signal processing approaches, such as wavelet domain optimized Savitzky-Golay filtering [36], Fourier-Bessel series expansion-empirical wavelet transform (FBSE-EWT) based rhythm decomposition, and adaptive empirical techniques, particularly targeting motion and ocular artifact removal [37, 38]. These methods have contributed to improving artifact suppression through time frequency domain integration. Many of these hybrid approaches, whether traditional or statistical, still rely on manual parameter tuning or threshold based heuristics, which may limit their scalability and generalizability across datasets [39, 40]. The development of deep learning (DL) has led to promising advances in EEG signal analysis. DL models can learn complex, nonlinear transformations from noisy to clean signals without relying on reference features [41]. Initial studies applied DL to classify artifact related components in ICA [42, 43]. Recent studies have focused on end-to-end denoising. Models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders (AE), generative adversarial networks (GANs), transformers, and hybrid networks have demonstrated significant performance in learning denoising transformations directly from raw EEG data [44].

This review explores the field of DL based EEG denoising, detailing model architectures, training strategies, evaluation metrics, and comparative performance. Key

challenges such as generalization, interpretability, and data availability are discussed, while highlighting emerging trends. It is important to note that deep learning models often exhibit a trade off between denoising performance and computational complexity. High performing models, such as transformer based or multi branch hybrid architectures typically offer superior artifact suppression at the cost of higher computational resources. This makes them less suitable for low latency or resource constrained environments. On the other hand, models such as shallow CNNs or simple autoencoders are computationally efficient and more appropriate for real time applications though they may yield lower denoising accuracy. Hence, this review considers this trade off while analyzing and comparing different model architectures, offering insights into their practical applicability.

A detailed literature search was conducted in accordance with PRISMA guidelines using databases including IEEE Xplore, PubMed, Scopus, Web of Science, and SpringerLink. The keywords used were: "EEG artifact removal", "EEG denoising", "deep learning models for EEG", "denoising", "CNN for EEG", "autoencoder for EEG", "RNN for EEG", "GAN for EEG", and "transformer based models for EEG". The search yielded a total of 712 records. After 132 duplicates were removed, 580 titles and abstracts were screened. A total of 234 full-text original research articles were reviewed for eligibility based on the following inclusion criteria: (1) application of deep learning for EEG artifact removal, (2) quantitative performance evaluation, and (3) availability of dataset or experimental validation. Following this, 205 studies were excluded due to a lack of deep learning relevance or inadequate methodology. Finally, 29 studies were included in this review for detailed analysis. The selection process is briefed in the PRISMA flow diagram (Fig. 1).

The key contributions of this review are as follows:

- Synthesizing 29 peer reviewed studies from 2020 to 2024 focused solely on deep learning based EEG denoising.

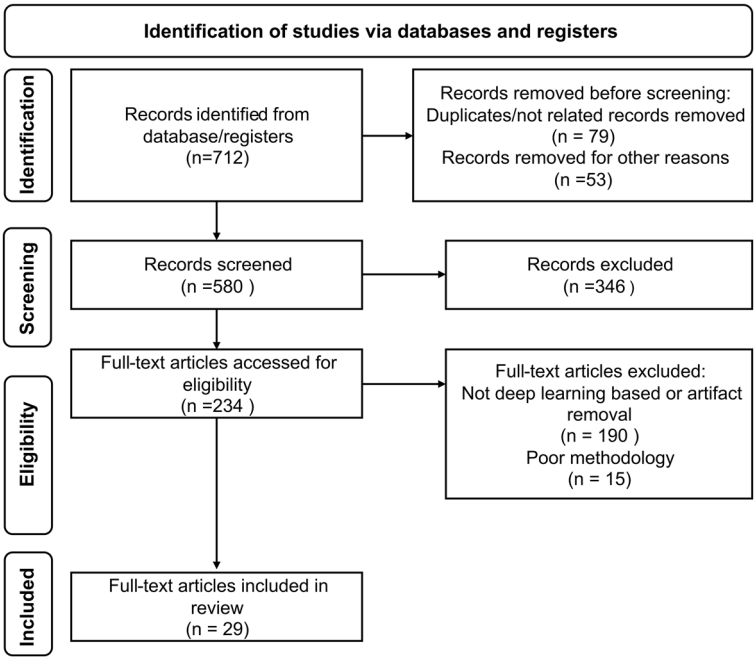



**Fig. 1** The PRISMA flow diagram

- Providing a structured comparison of CNN, autoencoder, GAN, and transformer models while detailing training strategies and evaluation pipeline to identify common practices and gaps.
- Highlighting the role of self-supervised learning, federated learning, and neuromorphic computing in overcoming existing challenges in EEG denoising.
- Discussing the trade offs between model performance, computational cost, and suitability for real time applications to guide both research and practical applications.

In line with these contributions, and following recommended practices for enhancing clarity in review studies [45], this review seeks to address the following research questions:

Q1. What architecture and training strategies have been used for EEG denoising, and how do they compare in performance, computational efficiency, and suitability for real time applications?
Q2. Which metrics and datasets are most effective for assessing artifact removal while preserving neural information?
Q3. What are the key challenges in current methods, and how might emerging approaches such as self-supervised, federated learning, and neuromorphic computing address them?

The structure of the Review is as follows: Sect. 2 explains the deep learning principles for denoising EEG signals. It also describes the advantages and limitations of various existing deep learning-based EEG denoising networks. Sections 3 through 6 provide details on the evaluation metrics and benchmarking, performance comparison of the networks, associated challenges, and emerging trends. Lastly, Sect. 7 concludes the review and outlines future directions. A comprehensive list of acronyms/abbreviations used in this article is listed in Table 1.

## 2 Deep learning for EEG denoising

Researchers are increasingly exploring advanced deep learning (DL) architectures for EEG signal denoising due to their capacity to learn complex, nonlinear mappings between noisy and clean signals. In denoising applications, a DL model is trained to approximate a nonlinear function $f_\theta$ that maps the noisy EEG signal $y$ to an estimate of the underlying clean signal $x$, where:

$$y = x + z \tag{1}$$

In Eq. (1), $z$ represents noise or artifact contamination. The DL network aims to minimize the divergence between the estimated clean signal $f_\theta(y)$ and the ground truth $x$ by optimizing the model parameters $\theta$, which include weights and biases. Generally, mean square error (MSE) is utilized as the loss function and is given by Eq. (2),

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}\left(f_\theta(y_i) - x_i\right)^2 \tag{2}$$

where $n$ denotes the number of samples of the segment and subscript $i$ represents the index of the sample. Optimization algorithms such as the RMSProp, ADAM, or Stochastic Gradient Descent (SGD) are typically used to reduce the loss function during training

**Table 1** List of abbreviations used in the article

| Acronym | Description |
|---|---|
| 1D Res-CNN | One Dimensional Residual Convolutional Neural Network |
| ACMD | Adaptive Chirp Mode Decomposition |
| ADAM | Adaptive Moment Estimation |
| AE | Autoencoder |
| ASR | Artifact Subspace Reconstruction |
| CAP | Cyclic Alternating Pattern |
| CC | Correlation Coefficient |
| CNN | Convolutional Neural Network |
| DAE | Deep Autoencoder |
| DARTS | Differentiable Architecture Search |
| DNN | Deep Neural Network |
| DTPNet | Densely-connected Temporal Pyramid Network |
| DuoCL | Dual-scale Convolutional neural network-Long short-term memory |
| DWT | Discrete Wavelet Transform |
| ECG | Electrocardiogram |
| EEGDNet | EEG Denoising Network |
| EEGIFNet | EEG Interactive Fusion Network |
| EMD | Empirical Mode Decomposition |
| EEMD | Ensemble Empirical Mode Decomposition |
| EMG | Electromyogram |
| FCNN | Fully Connected Neural Network |
| FICA | Fast Independent Component Analysis |
| FLNN | Functional Link Neural Network |
| FPN | Feature Pyramid Network |
| GAN | Generative Adversarial Network |
| GCTNet | GAN guided parallel CNN and Transformer Network |
| GPU | Graphics Processing Unit |
| GRU | Gated Recurrent Unit |
| HHT | Hilbert Huang Transform |
| JADE | Joint Approximate Diagonalization of Eigenmatrices |
| KL | Kullback-Leibler |
| kNN | K-Nearest Neighbor |
| LSTM | Long Short-Term Memory |
| MAE | Mean Absolute Error |
| MARA | Multiple Artifact Rejection Algorithm |
| MCC | Matthews Correlation Coefficient |
| MMNNet | Multi Module Neural Network |
| MSE | Mean Square Error |
| MTV | Multiresolution Total Variation |
| MWTV | Multiresolution Weighted Total Variation |
| PCA | Principal Component Analysis |
| PCDARTS | Partially-Connected Differentiable Architecture Search |
| PSD | Power Spectral Density |
| RBFN | Radial Basis Function Networks |
| RLS | Recursive Least Squares |
| ReLU | Rectified Linear Unit |
| RRMSE | Relative Root Mean Square Error |
| RMSProp | Root Mean Square Propagation |
| RNN | Recurrent Neural Network |
| SAE | Sparse AutoEncoder |
| SCNN | Simple Convolution Neural Network |
| SGD | Stochastic Gradient Descent |
| SM-EPO | Spider Monkey-based Electric Fish Optimization |

**Table 1** (continued)

| Acronym | Description |
|---|---|
| SNR | Signal to Noise Ratio |
| SOBI | Second Order Blind Identification |
| SuBAR | Surrogate Based Artifact Removal |
| STFNet | Spatial-Temporal Fusion Network |
| SVM | Support Vector Machine |
| TUH | Temple University Hospital |
| VAE | Variational Auto-Encoder |
| VFLN | Vector Functional Link Network |
| VMD | Variational Mode Decomposition |
| WGAN | Wasserstein Generative Adversarial Network |
| WOSG | Wavelet domain Optimized Savitzky–Golay filtering |
| WPD | Wavelet Packet Decomposition |
| WT | Wavelet Transform |
| WQN | Wavelet Quantile Normalization |

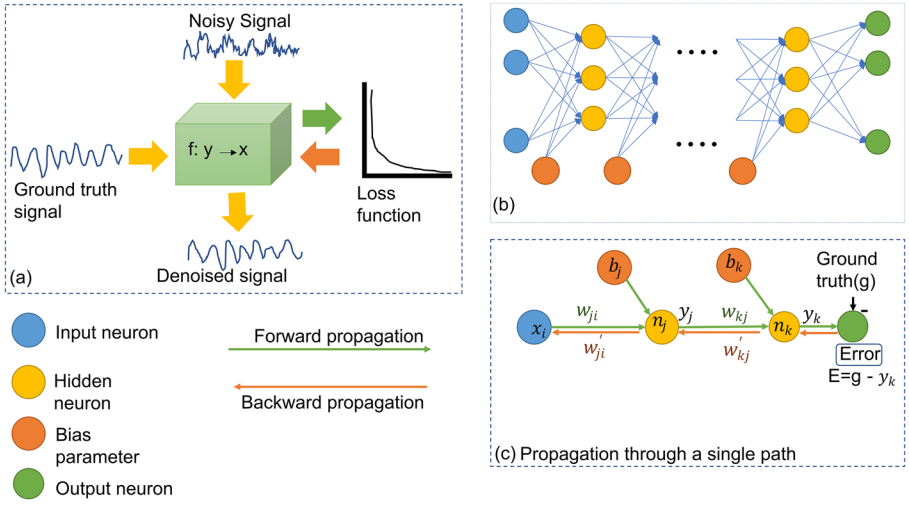



**Fig. 2** The basic principle of denoising using a deep learning network (**a**) represents the nonlinear function $f_\theta$ that maps the noisy EEG signal $y$ to an estimate of the underlying clean signal $x$, (**b**) is a simple, fully connected deep neural network, (**c**) illustrates the process of forward and backward propagation in a single path of deep neural network

[44]. The basic concept of DL based denoising is illustrated in Fig. 2, where the parameters $w$ and $b$ represent the weights and biases of the network, respectively. Weights control the contribution of each input to a neuron's output, whereas biases act as learnable offset terms. In each layer, the neuron activation is computed, passed forward, and the loss is backpropagated to update parameters via gradient descent.

During forward propagation, neuron activation is computed layer by layer. For a given hidden neuron $j$, the output is (as illustrated in Fig. 2(c)):

$$y_j = f\left(\sum_i w_{ji}x_i + b_j\right) \tag{3}$$

Similarly, for an output neuron $k$, the result becomes:

$$y_k = f\left(\sum_j w_{kj} y_j + b_k\right) \tag{4}$$

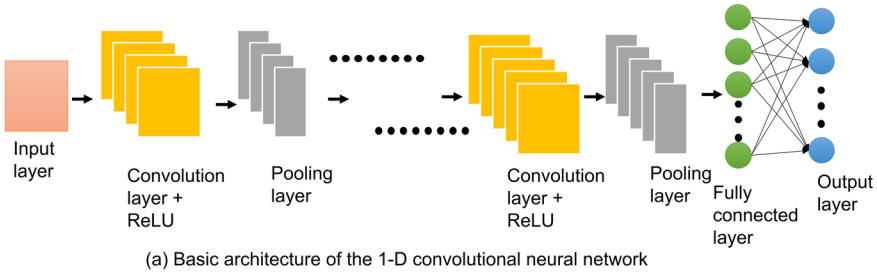
The output of each layer served as the input to the next layer until the final estimated signal was produced. After the forward pass, the network computes the error (loss function) and initiates backward propagation to update the weights and biases using gradient descent.

(For completeness, the full weight update equations for both output and hidden layers are provided in the Supplementary Material - Equations S1 to S4)

DL architectures can be broadly categorized based on the types of operations and layer configurations employed. Each architecture offers unique strengths in modeling the spatial, temporal, or contextual features of EEG signals. In the context of artifact removal, specific architectural choices influence a network's ability to generalize across subjects, preserve neural information, and suppress diverse artifact types. The following subsections provide a comprehensive overview of various deep learning models such as CNNs, RNNs, autoencoders, GANs, and transformer based models that have been utilized for EEG artifact removal.

### 2.1 Convolutional neural network (CNN)

Convolutional neural networks (CNNs) have emerged as a widely adopted architecture for EEG denoising because of their ability to automatically learn hierarchical features and extract the spatiotemporal structure of EEG signals [46]. The basic architecture of a CNN that is suitable for denoising tasks is illustrated in Fig. 3(a), while the detailed layer-wise parameters are summarized in Fig. 3(b). A typical CNN consists of multiple one- or two-dimensional convolutional layers interleaved with nonlinear activation functions (for instance, ReLU), pooling layers for dimensionality reduction, and fully connected layers for final regression or classification. In the case of EEG, one-dimensional convolutions are particularly effective as they model temporal patterns while reducing computational complexity. This is because EEG signals are inherently one-dimensional time series per channel, and 1D convolutions preserve this structure while enabling effective feature extraction along the time axis. In contrast, 2D CNNs require reshaping the EEG




(a) Basic architecture of the 1-D convolutional neural network

| Layer | Parameters to be defined |
|---|---|
| Convolution + ReLU | input_channels; output_channels; kernel_size; stride; padding |
| Pooling layer | kernel_size; stride (max_pooling / min_pooling / average_pooling) |
| Fully connected layer | Input_features; output_features |

(b) Essential parameters

**Fig. 3** A simple CNN architecture (**a**) basic architecture of a 1D CNN model, (**b**) Essential layer-wise parameters and configurations

signals into image-like formats, which can obscure temporal dependencies and introduce unnecessary computational burden. In Fig. 3, the first convolution layer often functions as a bandpass filter, isolating key frequency components, whereas deeper layers capture more abstract temporal dependencies. Thus, CNNs can be interpreted as end-to-end learnable filter banks that transform noisy EEG inputs into denoised outputs by optimizing the convolutional kernels.

Various enhancements have been proposed over the basic CNN architecture to improve the denoising performance. These include increasing the network depth [47], optimizing the number and configuration of hidden layers [48], incorporating inception style modules to capture multi scale features [49], and introducing custom blocks adapted to EEG characteristics [50]. Such architectural innovations have shown promise in improving the denoising performance and robustness to artifact variability, as summarized in Table 2. Early studies relied on segmenting fixed length EEG windows, tuning hyperparameters, and using loss functions for supervised learning. More recent work has focused on adaptive CNN architectures capable of handling EEG signals of varying lengths and learning nonlinear signal characteristics more effectively. Most CNN models are trained in a supervised setting by using clean and noisy EEG data.

While CNNs have proven effective, their performance is influenced by several architectural and training considerations (inferences from Table 2):

- Limited generalization: CNNs often exhibit reduced performance for unseen subjects or novel artifact types. Additionally, traditional 1D convolutions may not adequately capture spatial correlations across EEG channels.
- Smoothing effect of MSE: Standard training with MSE loss can yield smooth outputs, potentially eliminating important neural features. Recent studies have addressed this by combining spectral and temporal root mean square error (RMSE) loss functions to preserve signal integrity more effectively.
- Short-term dependencies: Conventional CNNs are limited in their ability to capture long term temporal dependencies. Solutions include the addition of stacked or dilated convolutions, or integrating CNNs with recurrent (RNN) or attention mechanisms to model long range patterns in EEG data.

### 2.2 Recurrent neural network (RNN)

Recurrent neural networks (RNNs) are designed to model sequential data with temporal correlations. Their ability to capture patterns over time makes them particularly useful for analyzing time series signals such as EEG signals. The fundamental architecture of an RNN is shown in Fig. 4. At each time step $\boldsymbol{t}$ (which varies from $[1,\ 1+\boldsymbol{t}]$), the network receives an input vector $\boldsymbol{x_t}$ and a hidden state $\boldsymbol{h_{t-1}}$ from the previous time step, which serves as a memory mechanism for capturing temporal dependencies. The output $\boldsymbol{y_t}$ and hidden state $\boldsymbol{h_t}$ are computed according to (9).

$$\begin{cases} \boldsymbol{h_t} = tanh(\boldsymbol{W_{hh}h_{t-1}} + \boldsymbol{W_{xh}x_t}) \\ \boldsymbol{y_t} = \boldsymbol{W_{hy}h_t} \end{cases} \tag{5}$$

One of the primary challenges of a basic RNN is the vanishing gradient problem, which limits its ability to learn long term dependencies. To address this, two variants - gated recurrent units (GRU) and long short-term memory (LSTM) have been introduced.

**Table 2** Summary of studies based on CNN for the denoising of EEG signals

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [49] | 2020 | 1D-ResCNN | Ocular, cardiac, muscle | CHB-MIT, Physionet | SNR, RMSE, CC (0.9567) | Kurtosis value and thresholding | Both | MSE/ADAM | Low | ICA, FICA, RLS filer, WT, DNN |
| [48] | 2021 | 1D- CNN + SM-EPO | Ocular, cardiac, muscle | CHB-MIT | MAE, PSNR, RMSE, SDR, CC (0.749), CSED | Hyper-parameter tuning | Single | PSNR + RMSE loss/ custom optimizer | High | FLM, Adaptive FLM–RBFN, SVM and DLN |
| [46] | 2021 | Deep CNN | Ocular, cardiac, muscle, electrode connection issues | Real EPILEPSIAE | PCC (0.86), RMSE, RRMSE, $SNR_{diff}$ | ICA decomposition | Multi | RRMSE/ADAM | High | 1D-ResCNN, extended Infomax ICA, MARA classifier |
| [47] | 2022 | MMNNet | Ocular, muscle | EEGdenoiseNet | RRMSE (temporal & spectral), CC (0.959, 0.896) | Residual connection + hyper parameter selection | Single | MSE/ADAM | Low | FCNN, Simple CNN, Complex CNN, RNN, and Novel CNN |
| [51] | 2023 | 1D-CNN | Motion | PhysioNet | MAE, ΔSNR, CC (0.905), Construction error | Spatial pooling and bilinear interpolation | Single | MSE/ADAM | High | FPN, LinkNet, UNet and its variants |
| [50] | 2024 | DTP-Net | Ocular, muscle | EEGdenoiseNet, Semi-Simulated EEG/EOG, BCI dataset, MNE M/EEG | ΔSNR, RRMSE, precision, recall, F1 scores, CC (0.926,0.9812) and Kappa statistics | Temporal pyramids | Both | MSE/ADAM | High | SimpleCNN, 1D-ResCNN, NovelCNN, WQN |

Both architectures use gating mechanisms to retain important information across time steps. GRUs are faster to train and require less data than LSTMs, making them a better choice when resources are limited.

Although RNNs have been broadly used in classification and sequence modeling tasks, their application to EEG artifact removal remains limited. McIntosh et al. [52] proposed an RNN-based model to suppress ballistocardiogram (BCG) artifacts in EEG/fMRI data by learning nonlinear mappings between contaminated EEG and reference ECG signals. While this model demonstrated real-time adaptability to physiological variations (for instance, heart rate), it assumed independence between the EEG and ECG signals. Such assumptions may not hold in practical applications, potentially reducing their effectiveness.

There are several limitations that must be considered when designing RNN-based EEG denoising models:

- High computational cost when handling multi-channel or high dimensional EEG data.
- Vanishing gradients and overfitting risks, specifically with long input sequences.
- Limited generalization to unseen subjects or noise types.
- Inability to simultaneously model spatial and temporal features, unless combined with other architectures.
- Sensitivity to hyperparameter tuning, which can significantly affect denoising performance.

Despite these challenges, RNNs, particularly when integrated with CNNs or autoencoders, can enhance temporal modeling and improve denoising efficacy [53–55]. For further architectural insights, readers may refer to [56], which provides a detailed discussion of RNNs, GRUs, and LSTMs.

### 2.3 Autoencoders

Autoencoders are powerful neural networks that learn to extract key features from data through encoding and decoding. In EEG denoising, they are used to reconstruct clean signals from noisy inputs by learning the efficient representations of the underlying signal structure. One of their key advantages is that they can be trained even without labeled data, making them useful for unsupervised learning tasks. A typical autoencoder architecture is shown in Fig. 5. It consists of three main parts:

- Encoder compresses the input into a low dimensional latent representation.
- Latent (bottleneck) layer captures essential signal features while filtering out noise.
- Decoder reconstructs the signal from the latent space.

Mathematically, if $\boldsymbol{X}$ is the input signal, the encoder transforms it according to Eq. (10):

$$\boldsymbol{h} = \boldsymbol{f}(\boldsymbol{X}) = \boldsymbol{f}(\boldsymbol{W}\boldsymbol{X} + \boldsymbol{b}_1) \tag{6}$$

$$\widetilde{\boldsymbol{X}} = \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{g}(\widetilde{\boldsymbol{W}}\ \boldsymbol{h} + \boldsymbol{b}_2) \tag{7}$$

Common types of autoencoders used for EEG denoising include: denoising autoencoders (DAE), variational autoencoders (VAE), sparse autoencoders, and deep convolutional autoencoders. These models have shown significant improvements in artifact removal

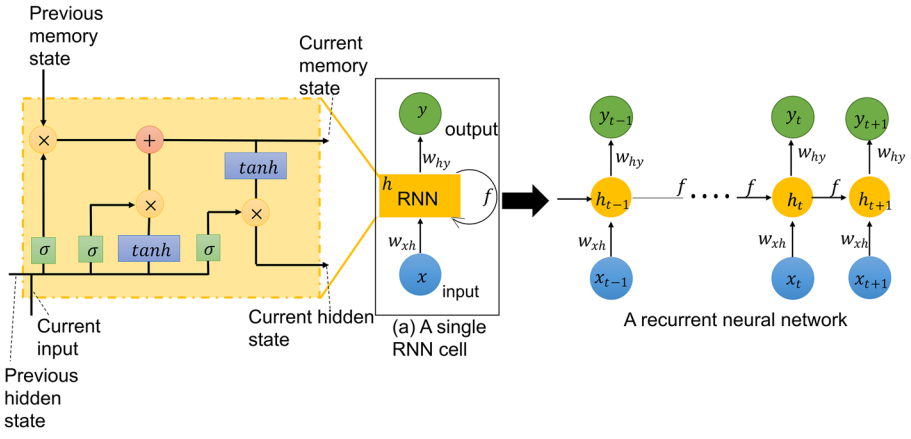



**Fig. 4** The basic architecture of an RNN model (**a**) A single RNN cell, overall RNN architecture is represented to the right of (**a**), inside each RNN cell is represented to the left of (**a**)

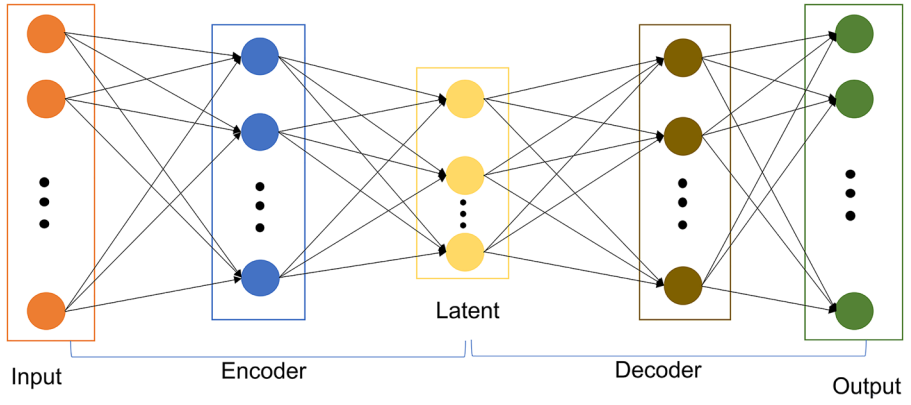



**Fig. 5** The basic architecture of an autoencoder network

and in enhancing EEG signal quality [57]. Deep or stacked autoencoders can model complex features and improve the reconstruction performance. However, an important design consideration, particularly in DAE, VAE, and sparse AEs, is the selection of latent space dimensionality. If selecting the latent space is too small, important neural features may be lost; if it is too large, noise may be preserved, or the model may overfit. This dimensionality is often determined empirically, which can be a challenge in the absence of ground truth. In addition, most autoencoder frameworks process 1D temporal features, which limits their ability to capture spatiotemporal patterns in EEG signals. This limitation can be addressed by integrating CNN or RNN modules within autoencoder structures [44, 54]. The mean squared error (MSE) between the reconstructed output and clean target is the most common loss function used (refer to Eq. 2).

In variational autoencoders (VAE), the loss function includes both MSE and Kullback-Leibler (KL) divergence terms, weighted by a factor $\beta$, to implement regularization [58]. Some advanced models also incorporate adversarial or composite loss functions to better capture nonlinear fluctuations in EEG signals [59]. A list of the autoencoder-based EEG denoising models is presented in Table 3. These studies show that autoencoders, particularly when combined with domain knowledge (for instance, hybrid loss functions and multi-path encoders), effectively suppress artifacts and improve signal quality.

**Table 3** Summary of studies based on autoencoder for the denoising of EEG signals

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/ Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [44] | 2018 | Stacked sparse AE | Ocular | BCI Competition IV | PSD, RMSE, Accuracy | Kurtosis value, Standardization of noisy EEG | Multi | MSE/ ADAM | Low | SAE, ICA, K-ICA, SOBI |
| [60] | 2020 | Wavelet + sparse encoder | Ocular | Visual attention task dataset, mental state recognition dataset, semi-simulated | PSD, Frequency correlation, MSE | Optimal wavelet decomposition | Single | Cost function/ bound-constrained optimization. | Low | ICA (SOBI, Infomax, JADE) |
| [54] | 2022 | GRU AE | Ocular, cardiac, muscle, motion | CAP (Physionet), MIT-BIH arrhythmia | Accuracy (98.52%), average CC (0.73), RRMSE, PSD | GRU network to identify artifacts and to extract features | Single | MSE + BCE/ AdamW | Low | SuBAR, EEMD-CCA |
| [59] | 2022 | U-Net + ICA | Ocuar, muscle, line noise | Experimental + simulation | MSR, SNR | ICA, ICLabel, tunable hyperparameters | Multi | Ensemble loss | High | ASR, 1D-ResCNN |
| [61] | 2022 | DeepSeparator | Ocular, muscle | EEGdenoiseNet, real EEG | RRMSE (temporal & spectral), CC (0.734) | Expert-annotated data | Both | MSE/ ADAM | Low | adaptive filter, HHT, EEMD-ICA |

**Table 3** (continued)

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/ Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [62] | 2024 | DAE | Ocular, motion, muscle | Semi-simulated, EEGdenoiseNet, Physionet | RRMSE (temporal & spectral), CC (0.85,0.70,0.79) | Multiple convolution layers | Single | MSE/ ADAM | Low | Simple CNN, RNN, Complex CNN, 1D Res-CNN, Novel CNN |

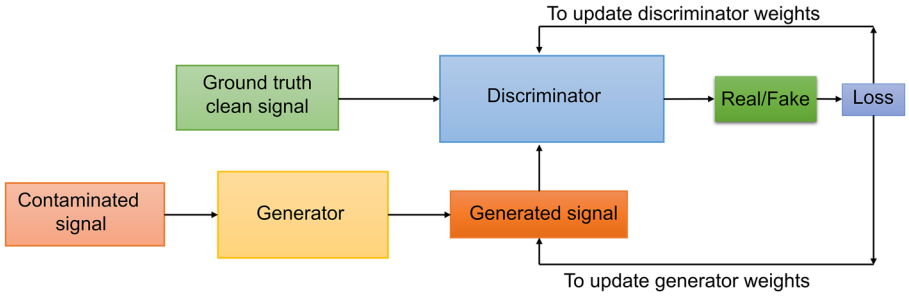



**Fig. 6** The basic architecture of a generative adversarial network

However, several challenges remain:

- Supervised models require clean datasets, whereas unsupervised models may overfit by learning to reproduce both the signal and noise (identity mapping).
- Autoencoders trained on a specific dataset may fail to generalize to different electrode configurations or artifact types.

### 2.4 Generative adversarial network (GAN)

Generative adversarial networks (GANs) have shown remarkable success in image processing tasks, including denoising, image synthesis, and artifact removal. Their application to time series data, such as EEG, is more recent and presents unique challenges. Initially introduced for data augmentation in EEG processing, GANs have been increasingly adopted for denoising tasks. A typical GAN architecture consists of two basic components: a generator ($\boldsymbol{G}$) and a discriminator ($\boldsymbol{D}$), as illustrated in Fig. 6. The generator is responsible for removing artifacts from the input EEG signal and reconstructing the clean version. The architectures for generators often include convolutional layers, autoencoder structures, or residual connections to better model the spatiotemporal features of the signal. The discriminator evaluates whether the denoised signal by the generator is indistinguishable from a clean EEG. Discriminators are typically built using convolutional layers and employ a custom loss function to facilitate the learning process.

The GAN training objective can be defined as:

$$\underset{D}{max}\underset{G}{min}\boldsymbol{E}_{\boldsymbol{X}\sim\boldsymbol{P}(\boldsymbol{X})}[\boldsymbol{log}\boldsymbol{D}\left(\boldsymbol{X}\right)]+\boldsymbol{X}_{\boldsymbol{X}^{\boldsymbol{raw}}\sim\boldsymbol{P}(\boldsymbol{X}^{\boldsymbol{raw}})}[\boldsymbol{log}(1-\boldsymbol{D}\left(\boldsymbol{G}\left(\boldsymbol{X}^{\boldsymbol{raw}}\right)\right))] \quad (8)$$

where $\boldsymbol{X}$ is the clean EEG, $\boldsymbol{X^{raw}}$ is the artifact contaminated input EEG and $\boldsymbol{P}(\boldsymbol{X})$ is the real data distribution.

GAN based EEG denoising models typically combine a reconstruction loss (for instance, MSE) with adversarial loss to ensure stable signal generation. Whereas preprocessing, such as normalization based on sample entropy or energy thresholds, is often essential for GAN stability during training [63]. The reconstruction loss for the GAN based denoising models can be expressed as:

$$\mathcal{L}_{GAN} = \mathcal{L}_{MSE} + \mathcal{L}_{adv} \tag{9}$$

where $\mathcal{L}_{MSE}$ is MSE loss (refer to Eq. 2) and $\mathcal{L}_{adv}$ is adversarial loss. The adversarial loss component can be expressed according to Eq. (14).

$$\mathcal{L}_{adv} = \sum -\boldsymbol{logD}\left(\boldsymbol{G}\left(\boldsymbol{X^{raw}}\right)\right) \tag{10}$$

GANs can automatically handle multichannel EEG denoising and have demonstrated the potential to generalize across different subjects and artifact types. Several GAN variants have been applied in this domain, as summarized in Table 4:

- Cycle GAN: Enables denoising without paired clean data using cycle consistency, making it well suited for unlabeled datasets [64].
- Wasserstein GAN (WGAN): Improved training stability using the Wasserstein distance as a loss function. WGAN [65].
- Spectral loss in GANs: Several GAN architectures incorporate additional loss function terms, such as spectral divergence, to more effectively maintain the signal energy and frequency content.

Although GAN based EEG denoising shows significant potential, it faces several challenges:

- Training instability: GANs are prone to oscillating losses and are sensitive to hyperparameters. WGANs address this problem using a smoother training objective.
- Mode collapse: Generators can produce repetitive outputs for varied inputs. This can be solved using a conditional GAN or multiple discriminators.
- Data limitations: Training requires a clean EEG as the ground truth, which is rarely available in real-world recordings. As a result, researchers often use semi-synthetic or augmented datasets. Although effective, the use of synthetic data raises ethical considerations regarding data authenticity and the potential for misleading interpretations, specifically in clinical applications. It is essential that synthetic data is transparently labeled, validated, and not overly relied upon.
- Computational cost: GAN training requires high computational resources, large datasets, and significant preprocessing.
- Preservation of neural information: Overfitting to artifact patterns or aggressive training can distort genuine EEG activity, which may limit the model's clinical utility.

**Table 4** Summary of studies based on GAN for the denoising of EEG signals

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [66] | 2022 | GAN | Ocular | EEG Eye Artifact data, BCI Competition IV 2b, Multimodal Signal Dataset | SNR, PCC (0.945), RRMSE (temporal & spectral), PSD | Normalization of input data, ground truth data | Multi | MSE + adversarial loss | Moderate | CNN based AE |
| [63] | 2022 | GAN | Ocular, muscle, cardiac, Power line noise | HaLT | CC (0.771), RMSE | Threshold parameters for data normalization, ground truth data, energy estimation | Multi | Earth-moving distance based/RMSProp | High | ICA, EMD, WT |
| [65] | 2023 | WGAN | Ocular, muscle | EEGdenoiseNet, experimental data | PSD, CC (0.726, 0.637), RRMSE (temporal & spectral), | Normalization of input data, ground truth data, Wasserstein distance | Multi | Wasserstein distance | High | Denoised autoencoder, Wiener filter, EMD |
| [64] | 2024 | CycleGAN | Motion | PhysioNet | SNR, Temporal correlation (0.88), spectral correlation (0.943), PSD | Self-attention blocks | Single | Adversarial loss + Cyclic consistency loss + identity loss/ADAM | High | DWT, EMD, EEMD, EMD-ICA/CCA, EEMD-ICA/CCA, SSA, DWT-MTV, DWT-MWTV, VMD-PCA/CCA, WPD-CCA |

### 2.5 Transformer network

Transform networks have a unique architecture, centered on the attention mechanism, which differentiates them from traditional deep learning networks such as CNNs and RNNs. The vanilla transformer architecture was originally proposed for natural language processing (NLP) applications and has recently been adapted for EEG signal processing, specifically for classification tasks and artifact removal. It can model long term temporal

dependencies and extract global features. In the context of EEG, the attention mechanism allows the model to focus on relevant time segments of the signal, helping to isolate artifacts that may not be localized or consistent in appearance. This is particularly useful when dealing with nonstationary signals such as EEG. The standard transformer structure [67], shown in Fig. 7, consists of stacked encoder and decoder blocks. Each encoder includes two key components: a multi-head self-attention block that focuses on different segments of the input and a feed-forward network (for instance, a fully connected layer).

A residual connection surrounds each sublayer, followed by layer normalization, which enhances the gradient flow during the training. Each decoder further includes a masked multi-head attention block to ensure nonlinear output processing, as shown in Fig. 7. To preserve the sequence information, positional encoding is added to the input embedding, because transformers do not intrinsically model sequences. Generally, positional encoding involves sine and cosine functions of different frequencies to embed location information according to (15) and (16).

$$\boldsymbol{PE}_{(\boldsymbol{pos},2\boldsymbol{i})} = \sin\frac{\boldsymbol{pos}}{10000^{\,2\boldsymbol{i}\,/\,\boldsymbol{d}_{\boldsymbol{model}}}} \tag{11}$$

$$\boldsymbol{PE}_{(\boldsymbol{pos},2\boldsymbol{i}+1)} = \cos\frac{\boldsymbol{pos}}{10000^{\,2\boldsymbol{i}\,/\,\boldsymbol{d}_{\boldsymbol{model}}}} \tag{12}$$

where, $\boldsymbol{pos}$ is the position index, $\boldsymbol{i}$ is the embedding dimension index and $\boldsymbol{d_model}$ is the total number embedding dimensions. Prior to entering the encoder, the raw EEG input $\boldsymbol{X}$ is mapped to a high dimensional space through a feature embedded representation ($\boldsymbol{E}$) according to (17).

$$\boldsymbol{E} = \boldsymbol{X}\,\boldsymbol{W}_{\boldsymbol{e}} + \boldsymbol{b}_{\boldsymbol{e}} \tag{13}$$

where $\boldsymbol{W}_{\boldsymbol{e}}$, and $\boldsymbol{b}_{\boldsymbol{e}}$ are the trainable projection parameters. The embedded inputs and positional encodings are then combined and passed into the encoder stack. The attention

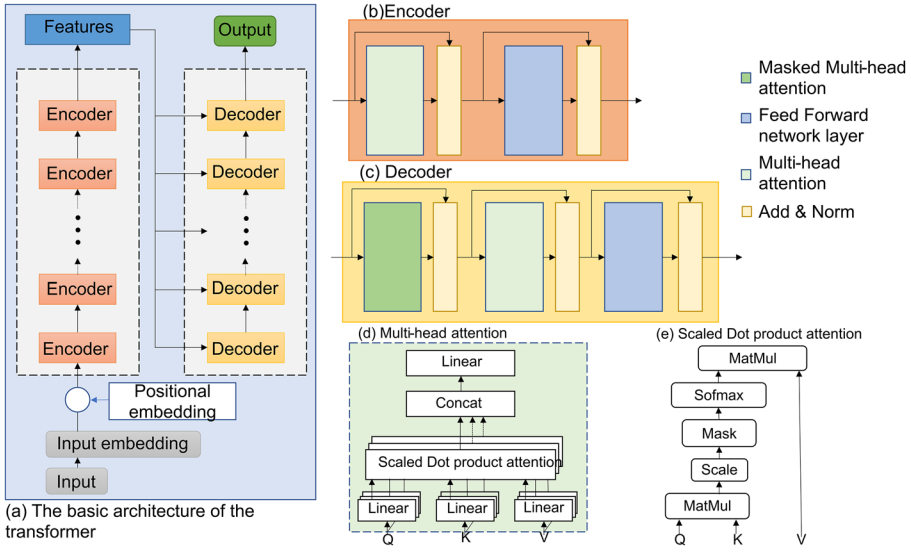



**Fig. 7** (**a**) The basic architecture of the transformer network, (**b**) Encoder block, (**c**) Decoder block, (**d**) architecture of a multi-head attention block, (**e**) illustration of scaled dot product attention mechanism

mechanism allows the model to focus on different parts of the data by assigning attention weights. For each attention head, the scaled dot product attention was computed according to (18).

$$\boldsymbol{Attention}\,(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \boldsymbol{softmax}\left(\frac{\boldsymbol{QK^T}}{\sqrt{\boldsymbol{d_k}}}\right)\boldsymbol{V} \tag{14}$$

Each attention head $\boldsymbol{H_i}$ is derived as:

$$\boldsymbol{H_i} = \boldsymbol{Attention}\left(\boldsymbol{QW_i^Q}, \boldsymbol{KW_i^K}, \boldsymbol{VW_i^V}\right), \quad 1 \le \boldsymbol{i} \le \boldsymbol{h} \tag{15}$$

Multiple heads are concatenated and projected again to reconstruct the feature space according to (20).

$$\boldsymbol{MultiHead}\,(\boldsymbol{Q},\ \boldsymbol{K},\boldsymbol{V}) = \boldsymbol{Concat}\,(\boldsymbol{H}_1,\boldsymbol{H}_2,\boldsymbol{H}_{3,}\ldots.,\boldsymbol{H_h})\,\boldsymbol{W^o} \tag{16}$$

where $\boldsymbol{Q},\ \boldsymbol{K},\boldsymbol{V}$ are derived from positional encoding: $\boldsymbol{Q} = \boldsymbol{PE_{pos}W^Q}$, $\boldsymbol{K} = \boldsymbol{PE_{pos}W^K}$, and $\boldsymbol{V} = \boldsymbol{PE_{pos}W^V}$. $\boldsymbol{W^Q}$, $\boldsymbol{W^K}$, $\boldsymbol{W^V} \in$ R $(\boldsymbol{d_{model}}\ \boldsymbol{X}\ \boldsymbol{d_{model}}/\boldsymbol{h})$. $\boldsymbol{h}$ denotes the number of attention heads, and $\boldsymbol{W^o}$ represents the weight matrix for the corresponding linear mapping. The feed forward network within each encoder and decoder applies two linear transformations with ReLU activation according to (21)

$$\boldsymbol{FFN}\,(\boldsymbol{x}) = \mathbf{max}\,(0, \boldsymbol{x}\boldsymbol{W}_1 + \boldsymbol{b}_1)\,\boldsymbol{W}_2 + \boldsymbol{b}_2 \tag{17}$$

where $\boldsymbol{W}_1$, $\boldsymbol{W}_2$, $\boldsymbol{b}_1$, and $\boldsymbol{b}_2$ are the weight and bias parameters.

Different variations of transformer architectures have been explored for EEG denoising.

- Time series transformers replace standard input embeddings with windowed representations of EEG sequences [68].
- Vision transformers treat time frequency spectrograms or topographic EEG maps as image like inputs. These transformers are primarily used for classification tasks, and a similar approach can be extended to denoising.
- Spectral transformers operate on transformed representations such as the Fourier or wavelet spectra of EEG signals.

In most applications, the transformer is trained as an encoder only or an encoder-decoder model that maps artifact contaminated EEG to clean outputs [69]. An overview of transformer based EEG denoising studies is summarized in Table 5.

The challenges associated with transformer based EEG denoising models are listed as follows:

- Data requirements: Transformers often need extensive training data or augmentation to generalize effectively.
- Generalization: Models trained on one dataset may not perform well with data from different populations or electrode montages. Cross-subject learning and domain adaptation may help, but can be complex.

**Table 5** Summary of studies based on transformer for the denoising of EEG signals

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [69] | 2022 | EEGDNet | Ocular, muscle | EEGdenoiseNet | RRMSE (temporal & spectral), CC (0.868,0.732) | Transforming 1D signal into 2D form | Single | MSE/ADAM | Low | DLN, SCNN, RNN, 1D-ResCNN |
| [68] | 2024 | Denosieformer | Ocular, muscle | EEGdenoiseNet, experimental, semisynthetic | RRMSE, CC (0.930) | Slice pattern attention module | Single | MSE + Kl divergence/AdamW | Medium | VAE, RNN-DNN, GAN, Complex-CNN, Novel CNN, DuoCL, EEGDNet |

- Although transformers model long term dependencies, they may miss fine grained details when handling long EEG sequences. Techniques such as downsampling or convolutions can shorten the input length, but risk information loss.
- Computational cost: The self attention mechanism requires significant amount of resources. Multichannel EEG denoising with transformers requires a large GPU memory and training time.
- Owing to their depth and complexity, transformer models are more difficult to interpret than traditional models.

### 2.6 Hybrid deep learning models

Hybrid deep learning models have demonstrated notable success in EEG artifact removal by combining the strengths of multiple neural network architectures. These models integrate diverse learning patterns spanning both the temporal and frequency domains to enhance the precision and adaptability of denoising models. Hybrid approaches often include combinations such as: CNNs with multi-head self-attention modules [68, 69], autoencoders embedded in GAN based generators, and architectures that simultaneously influence time frequency representations [70–72]. Compared with conventional DL models, hybrid architectures can better handle cross-domain knowledge and complex noise patterns.

For instance, GCTNet is a model that fuses CNN and transformer modules in parallel with a GAN network. The generator captures both the local and global temporal dependencies using spatial-spectral attention mechanisms. This combination enables the model to exploit its inter channel and spectral aspects more effectively. Other studies have incorporated channel attention mechanisms to selectively enhance informative EEG channels while suppressing less relevant channels. In addition, self-attention modules have been used to model inter channel correlations, improving denoising performance across multi-channel processes. A summary of recent hybrid deep learning models used for EEG denoising is presented in Table 6.

Key advantages of hybrid architectures (inferred from Table 6):

**Table 6** Summary of studies based on hybrid models for the denoising of EEG signals

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [73] | 2022 | Deep Wavelet- VFLN | Ocular | Mendeley database | MSE (0.20), RMSE | Reference EEG, parameter estimation | Multi | Sum square error/RLS algorithm | Low | - |
| [72] | 2022 | Dual domain CNN | Ocular | EEGdenoiseNet, Semisimulated, CHB-MIT | SNR, CC, RRMSE | Concatenation of dual domain features | Single and a few channel | MSE/ADAM | Low | FCNN, Simple CNN, ResCNN |
| [74] | 2023 | kNN-LSTM | Ocular, muscle, | Experimental | CC(0.72, SAR, MAE, NMSE, SSIM | Accurate detection of artifacts | Both | Customized loss/ADAM | Moderate | EEMD-CCA, ICA-Wavelet, Tensor based |
| [75] | 2023 | ResU-net1D + SCNN, RNN + ResCNN | Ocular, muscle, motion | EEGdenoiseNet, CHB-MIT | RRMSE, CC (0,0.906) SNR | Artifact position mask generated by the segmentation network | Single | Binary cross entropy + MSE/ADAM | High | EEMD-ICA, EEMD-CCA, VMD-MA, VMD-OA, ACMD-OA |
| [76] | 2023 | Uncertainty-aware Net | Ocular, muscle | ISRUC-S3, TUSZ | Change in SNR, NMSE, Change in correlation (0.84), Spectral coherence | Fused uncertainty parameters | Multi | Customized loss/ADAM | Low | EMD-ICA, EMD-CCA, WT, WQN, WOSG, DBPEWT, MMNN, Novel CNN,1D-ResCNN, DeepSeparator, GRUMARSC |
| [55] | 2023 | CNN-LSTM (DuoCL) | Ocular, muscle, cardiac, motion, baseline wander | Semi simulated, KLADOS | SNR, RRMSE (temporal & spectral), CC (0.94,099,0.99,0.964) | Morphological feature extraction with self-correlation and time frequency analysis | Both | MSE/ADAM | Low | FCNN, LSTM, 1D-Res CNN, Novel CNN |
| [71] | 2024 | GCTNet | Ocular, muscle | EEGdenoiseNet, Semi-simulated EEG/EOG dataset, Real EEG | RRMSE, SNR, CC, WCC, WSNR, PSD | Hyperparameter that regulates the loss functions | Both | MSE/ADAM | High | FCNN, SimpleCNN, ResCNN, Novel CNN, EEGDNet, DuoCL |

**Table 6** (continued)

| Article | Year | Model | Artifacts | Dataset used | Performance metrics | Dependency | Input type (channel) | Loss/Optimizer | Complexity | Benchmarked against |
|---|---|---|---|---|---|---|---|---|---|---|
| [77] | 2024 | EEGIFNet | Ocular, muscle, cardiac, motion, power line interference | EEGdenoiseNet, experimental data | RRMSE, CC (0.958,0.929), SNR | Interaction module for information communication | Single | MSE/ADAM | Medium | FCNN, SimpleCNN, RNN,1D-ResCNN, Novel CNN |
| [78] | 2024 | NN architecture search | Ocular, muscle, cardiac, motion | EEGdenoiseNet, MIT-BIH arrhythmia dataset, motion artifact dataset | CC (0.95), RRMSE, SNR | Large convolutional kernel | Single | MSE/ADAM | High | FCNN, SimpleCNN, ResCNN, Novel CNN, EEGDNet, DuoCL, EEGIFNet, DARTS, PCDARTS, β-DARTS |
| [70] | 2024 | STFNet | Ocular, muscle | Semi-simulated EEG/EOG dataset, Semi-simulated EEG/EMG, BCI 2a, BCI 2b, MNE Sample | RRMSE, CC (0.988), SNR | Efficiency of feature extractor modules | Multi-channel | MSE/ADAM | Low | ICA, CCA, EEMD-CCA, SimpleCNN, ResCNN, EEGDNet, DuoCL, GCTNet, EEGIFNet, EEGANet |

- Effective for both single-channel and multi-channel EEG signals denoising tasks.
- Can simultaneously capture short term temporal patterns, spatial features, and long term dependencies.
- Combining different architectures can lead to better performance across varied datasets.
- Hybrid models allow for custom loss function designs (for instance, combined MSE, adversarial, and spectral loss) that improve robustness.

Although hybrid architecture offers performance advantages, it also presents several design and computational challenges. A higher model complexity may result in more expensive training and more cautious tuning of the hyperparameters. Moreover, choosing an appropriate combination of architectures is essential for achieving a balance between computational efficiency, real-time applicability, and denoising accuracy. With EEG applications directed towards embedded and wearable platforms, developing hybrid models that are computationally efficient and powerful is a promising yet challenging direction in the field.

To consolidate the comparative insights from Tables 2, 3, 4, 5, 6 and 7 summarizes the strengths and limitations of several key representative studies. This synthesis highlights the diversity in architectural choices, the trade offs between denoising performances and computational efficiency, and the practical considerations. Such a summary enables a quick assessment of how different design strategies align with specific EEG denoising requirements, providing a useful reference for researchers before exploring emerging trends.

**Table 7** Summary of strengths and limitations of key studies

| Model/Study | Strengths | Limitations |
|---|---|---|
| 1D- ResCNN [49] | Fast computation, effective spatial feature extraction, low latency | Weak in modeling long term temporal dependencies, possible overfitting on small datasets |
| GRU autoencoder [54] | Captures temporal dependencies in non-stationary signals, robust to sequential noise patterns | Computationally intensive; harder to parallelize compared to CNNs |
| GAN [63] | Generates realistic artifacts-free EEG; strong generalization to unseen data | Training instability, requires large amounts of paired data |
| WGAN [65] | Improved stability over basic GAN, better spectral preservation | Increased training time, needs careful hyperparameter tuning |
| EEGDNet [69] | Models long range dependencies through attention mechanism, flexible input handling | High computational complexity; requires large training datasets |
| Denosieformer [68] | Efficient attention through slice patterns, strong temporal spectral denoising | Medium to high computational load; lack of generalization on real time applications |
| Dual domain CNN [72] | Employs time frequency features for improved accuracy | May struggle with complex motion artifacts |
| ResUNet 1D + SCNN/ RNN + ResCNN [75] | Strong artifact suppression, preserves signal morphology | Requires accurate artifact position masks; high complexity |
| Uncertainty aware Net [76] | Models uncertainty for better generalization; robust to diverse artifact types | May require complex calibration, additional computational steps |
| STFNet [70] | Highly accurate feature extraction, strong across multiple datasets | Model complexity may hinder real time applications |

### 2.7 Transfer learning and pretrained models in EEG denoising

Transfer learning is a strategy that involves re-purposing a model trained on one task for application to a related task, with minimal retraining. In EEG signal processing, transfer learning has been widely adopted in domains such as mental task classification, motor imagery classification and recognition, and artifact removal. In the case of EEG denoising, transfer learning generally involves pretraining deep learning models on large-scale synthetic or semi-synthetic datasets, where clean and noisy signal pairs are readily available. The pretrained models were then fine-tuned on smaller real-world EEG datasets, enabling them to adapt to real signal distributions while retaining robust noise suppression capabilities learned during pretraining. This approach helps to address the shortage of labeled EEG data and improves denoising performance. Another promising approach involves cross subject transfer, in which a model trained on EEG data from one group of subjects is adapted to new subjects without extensive retraining. This has significant potential in clinical or wearable EEG applications, where subject training may not be feasible. Techniques such as freezing early layers (which capture generic features), fine tuning deeper task specific layers and domain adaptation strategies are often used to improve generalizability while minimizing computational overhead.

Models such as ResNet [49], UNet [59], and Transformer-based architectures [68, 69], originally developed for image and sequence modelling, have been adopted and fine-tuned for EEG denoising tasks. Although transfer learning is advantageous, it presents several challenges.

Challenges in Transfer learning for EEG denoising:

- Mismatches between the source (pretraining) and target domains can lead to negative transfer where the performance in the target task degrades.
- Fine tuning on small real-world datasets may cause overfitting, specifically if domain gaps are large.
- Although transfer learning reduces training time, it still requires careful layer wise adaptation to balance performance and efficiency.

Despite these challenges, transfer learning remains a powerful tool in EEG denoising, particularly when annotated data are limited. Its success depends on the careful selection of source tasks, alignment of feature distribution, and proper fine tuning strategies suited to the target application.

## 3 Evaluation metrics and benchmarking

The effectiveness of an EEG denoising model is determined by its ability to suppress artifacts while preserving the underlying neural activity. Therefore, the choice of evaluation metrics plays a critical role in validating model performance. These metrics help distinguish between models that merely generate random outputs from those that meaningfully reconstruct artifact free, information preserving EEG signals. Commonly employed quantitative metrics include: signal-to-noise ratio (SNR), correlation coefficient (CC), root mean square error (RMSE), relative RMSE in temporal and spectral domains, and power spectral density (PSD). These metrics are typically computed between the model's reconstructed output and the ground truth clean EEG signal. The SNR helps to ensure the quality of the reconstructed signal; CC can be utilized to estimate the ability of the model to remove artifacts as well as its ability to preserve brain activities, depending on

the ground truth data available (ranges between 0 (min.) to 1 (max.)). The RMSE ensures that the developed model preserves the actual neural information.

A lower RMSE value indicates a better ability of the model to preserve the brain activity. The PSD plots indicate the presence and absence of certain frequency ranges related to brain activity and artifacts, respectively. Altogether, they provide a comprehensive view of the model's ability to denoise without distorting meaningful brain activity. A summary of the widely used evaluation metrics is presented in Table 8. In addition to quantitative metrics, visual tools such as PSD plots, time domain waveform comparisons, and time frequency representations (spectrograms) offer qualitative insights into the effectiveness of artifact suppression. While other metrics such as mutual information and spectral coherence, can offer insights into frequency domain coherence and information preservation, they are more suitable for connectivity analysis or specific spectral investigations. In contrast, SNR, CC, and RRMSE directly quantify artifact suppression and signal fidelity in both time and frequency domains, making them relevant for benchmarking EEG denoising performance.

Benchmarking establishes a standardized framework for comparing different EEG denoising models. Effective benchmarking involves three key components, namely dataset selection, comparative models, and validation. Dataset selection is fundamental in benchmarking. Researchers typically employ open-source datasets such as the PhysioNet EEG database (https://www.physionet.org/), Temple University Hospital (TUH) artifact corpus (https://isip.piconepress.com/), Kaggle database (https://www.kaggle.com/), sim ulated datasets, and semi-synthetic dataset EEGdenoiseNet [53]. Newly developed models are typically evaluated against baseline methods such as ICA, CCA, and existing deep learning models. Reporting improvements over these baselines demonstrates the added value of the proposed method. Finally, proper validation ensured the generalizability of the results. This includes subject-wise cross-validation, reporting the mean ± standard deviation across multiple subjects or trials. The overall workflow for model evaluation, including metric calculation, benchmarking, and result reporting, is illustrated in Fig. 8. This structured approach helps to ensure consistency and transparency in EEG denoising research, making comparative results more reliable and reproducible.Ð_.

**Table 8** Description of the most commonly used evaluation metrics

| Metric | It measures | Importance | Formula (x: ground truth, $\widetilde{x}$: reconstructed signal) |
|---|---|---|---|
| Signal-to-noise ratio | Ratio of clean signal power to noise power | Indicates the signal quality and direct measure of noise suppression effectiveness | $10\log_{10}\frac{\sum_{i=1}^{N} x_i^2}{\sum_{i=1}^{N}(\widetilde{x}_i - x_i)^2}$ |
| Temporal RRMSE | Error in time domain signal reconstruction | Assess the signal distortion | $\frac{RMS(\widetilde{x}-x)}{RMS(x)}$ |
| Spectral RRMSE | Error in the frequency domain | Evaluates the preservation of frequency components | $\frac{RMS(PSD(\widetilde{x})-PSD(x)}{RMS(PSD(x))}$ |
| Correlation Coefficients | Linear similarity between clean and denoised signals | Measures the shape preservation and signal integrity | $\frac{Cov(\widetilde{x},x)}{\sqrt{Var(\widetilde{x})\,Var(x)}}$ |
| Power spectral density | Energy is preserved in different frequencies and deviations in frequency content | Critical for applications relying on EEG frequency bands | $\frac{1}{N}\left\|\sum_{n=0}^{N-1} x[n]\, e^{-j2\pi\, fn/N}\right\|^2$ or Welch's method |

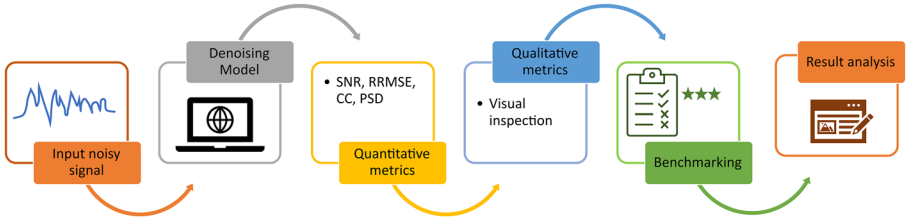



**Fig. 8** The overall workflow of evaluating EEG denoising models

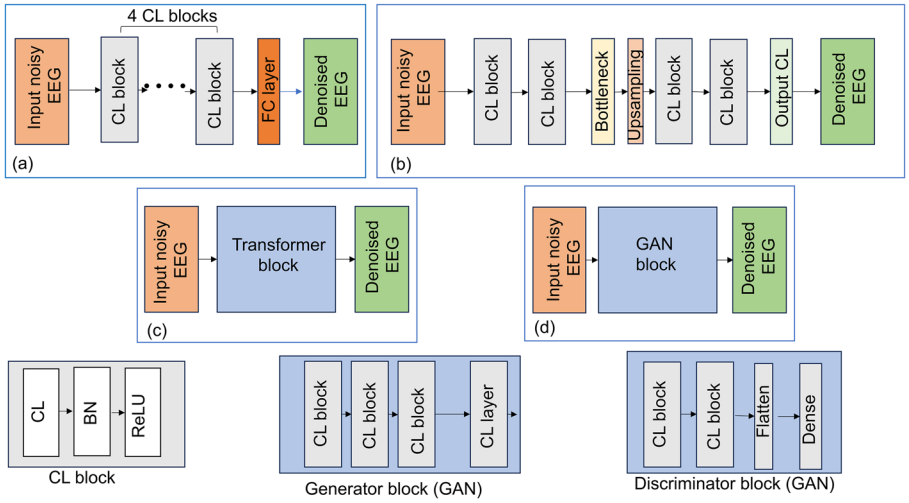



**Fig. 9** The model architecture implemented for performance comparison, (**a**) CNN based, (**b**) Autoencoder based, (**c**) Transformer based, (**d**) GAN based EEG denoising models. (CL: convolutional layer, FC: fully connected, BN: batch normalization, Transformer block and GAN block architectures are illustrated in Figs. 6 and 7, respectively)

## 4 Performance analysis of different DL models in denoising EEG signals

To evaluate the practical effectiveness of different deep learning (DL) architectures for EEG denoising, this study examined four prominent model types -CNN, autoencoder, GAN, and transformer under two artifact cases: ocular and muscle contamination. These baseline models were developed specifically for this study and were trained and tested under identical conditions using the EEGdenoiseNet dataset with a learning rate of 4.5e-5, the ADAM optimizer, and MSE as the loss function. It is important to note that RNNs were not included in this comparative analysis because of their significantly higher computational requirements and training complexity. Although RNNs (including LSTM and GRU variants) are capable of modeling temporal dependencies in EEG signals, they operate sequentially, making them integrally less parallelizable and slower to train compared to CNN or transformers. In addition, tuning RNNs often requires more extensive experimentation and resources. These constraints made it challenging to implement and evaluate RNNs alongside other architectures under a unified framework. Furthermore, to ensure consistency and isolate the core performance behavior, all the models were implemented in their simplest and most basic architectures, as shown in Fig. 9, without incorporating advanced modules such as attention fusion, residual connections, or domain-specific features. The results are presented in terms of the SNR, CC, and RRMSE as discussed in Sect. 3 (refer to Table 9). In addition, computational

**Table 9** Performance of different models in EOG (ocular) and EMG (muscle) artifact removal

| EOG artifact removal | | | | EMG artifact removal | | |
|---|---|---|---|---|---|---|
| **Model type** | **SNR (dB)** | **CC** | **RRMSE** | **SNR (dB)** | **CC** | **RRMSE** |
| **CNN** | **8.11** | **0.9194** | **0.3933** | 2.97 | 0.7721 | 0.7100 |
| **Autoencoder** | 5.50 | 0.8477 | 0.5310 | **4.42** | **0.7994** | **0.6013** |
| **GAN** | 2.92 | 0.7002 | 0.7143 | 3.53 | 0.7466 | 0.6662 |
| **Transformer** | 3.51 | 0.7444 | 0.7444 | 4.00 | 0.7762 | 0.6309 |

Note: The metrics provided correspond to the denoised EEG signal

**Table 10** Training and computational profile of different models in EOG (ocular) and EMG (muscle) artifact removal

| EOG artifact removal | | | | | EMG artifact removal | | | |
|---|---|---|---|---|---|---|---|---|
| **Model type** | **Training duration** | **Total parameters** | **Memory used** | **Training epochs** | **Training duration** | **Total parameters** | **Memory used** | **Training epochs** |
| **CNN** | 43 min | 16,816,064 | 64.15 MB | 39 | 24 min | 16,816,064 | 64.15 MB | 13 |
| **Autoencoder** | 28 min | 16,657 | 65.07 KB | 40 | 34 min | 16,657 | 65.07 KB | 30 |
| **GAN** | 3 h | (13,185 + 20,801) * | 51.50 KB + 81.25 KB | 40 | 4 h 15 min | (13,185 + 20,801) * | 51.50 KB + 81.25 KB | 30 |
| **Transformer** | 59 min | 141,569 | 553.00 KB | 50 | 55 min | 141,569 | 553.00 KB | 30 |

*including both generator and discriminator parameters

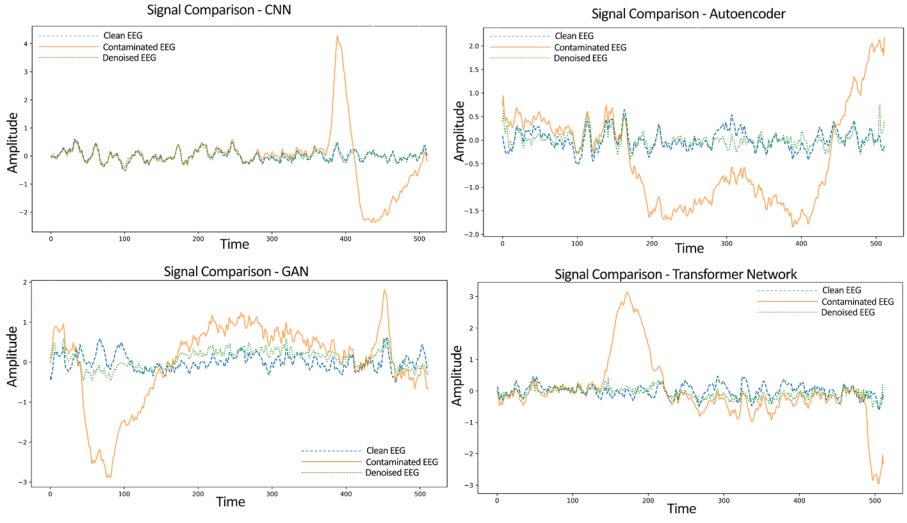



**Fig. 10** Time domain waveform comparison between clean, contaminated, and denoised EEG signals for EOG (ocular) artifact removal across different models

metrics such as the training duration, model size, and memory used are discussed (refer to Table 10).

Furthermore, signal comparison plots between clean (ground truth), contaminated (noisy), and denoised EEG signals demonstrate the qualitative improvements provided by each model, as shown in Figs. 10 and 11. These comparisons help to visualize artifact

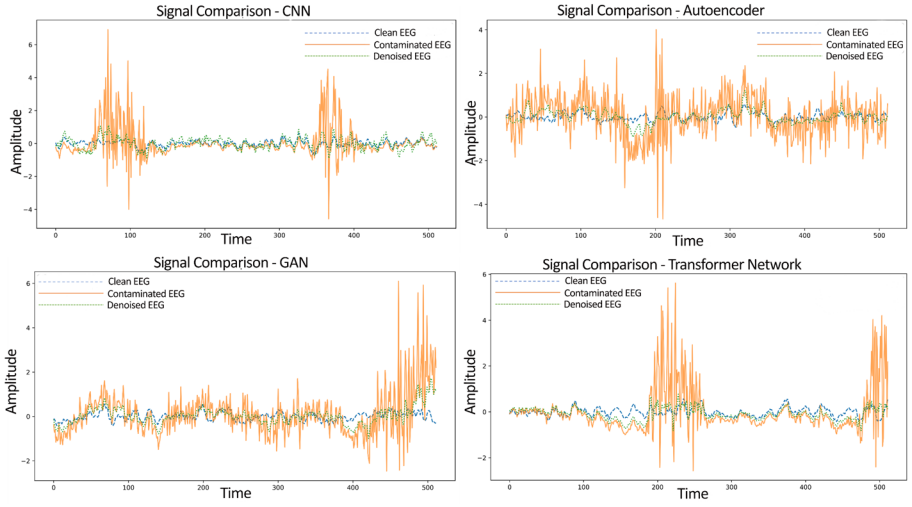



**Fig. 11** Time domain waveform comparison between clean, contaminated, and denoised EEG signals for EMG (muscle) artifact removal across different models

removal effectiveness and the preservation of underlying neural patterns. It is evident from the performance comparison metric that,

- CNN model consistently performed best across all metrics for EOG artifact removal, demonstrating a high SNR and strong CC values. Their large number of parameters contributed to better feature extraction, specifically for frontal EEG channels affected by ocular artifacts. However, this comes at the cost of increased memory and computational loads.
- Autoencoder offered a good trade-off between denoising quality and efficiency. With minimal parameters and fast training, they are well suited for embedded or wearable systems. However, their denoising performance was slightly lower than that of CNNs or transformers, particularly for ocular artifacts.
- The GAN demonstrated a relatively low denoising performance and required the longest training duration. Mode collapse and sensitivity to hyperparameters may have affected the performance.
- The transformer showed stronger performance for EMG artifact removal, achieving the highest SNR while maintaining moderate RRMSE. Their ability to model long range dependencies may help capture muscle induced distortions. However, they require higher memory usage and a relatively longer training time compared to CNN and autoencoders.

This performance comparison of different DL models suggests that each model offers distinct strengths and weaknesses that can vary depending on the specific use case. Hence, research should focus on combining these architectures to achieve a balance between denoising accuracy, generalizability, and computational efficiency for a target application (Additional details on data preparation and algorithm are provided in supplementary material).

## 5 Challenges and limitations

Despite recent advances in deep learning for EEG denoising, several key challenges continue to limit the performance, generalizability, and clinical translation of these models. A fundamental issue is the lack of clean ground truth data. Most real-world EEG recordings are contaminated with overlapping artifacts, which makes it difficult to obtain reliable reference signals for supervised learning. This scarcity has forced many studies to rely on semi-synthetic datasets, which may not fully capture the complexity of background EEG patterns. Furthermore, while many models achieve high performance on controlled or synthetic datasets, relatively few have been systematically evaluated real world EEG recordings with diverse artifact sources. Such cases often introduce nonstationary and mixed noise patterns, which can substantially degrade model generalization. In the absence of clean reference signals, alternative task driven evaluation strategies can be used to indirectly assess denoising performance. For instance, motor imagery classification accuracy can be compared before and after denoising using different methods. Improvements in classification accuracy can serve as an objective for effective artifact removal while preserving relevant neural features. Such approaches, however, are not yet consistently adopted in EEG denoising research.

Inter-subject and intra-subject variability in EEG signals present another significant hurdle. Neural activity patterns can vary considerably across individuals and even within the same individual over time, depending on the cognitive state, exhaustion, or sensor placement. These variations hinder the generalizability of models trained on limited datasets. The nonlinear and overlapping nature of artifacts such as EOG and EMG further complicates the denoising process. Without careful modeling, deep learning networks may risk removing relevant neural components along with artifacts, especially when noise frequencies overlap with the neural bands. Another limitation is the relatively small size of available EEG denoising datasets. The restricted amount of paired clean-noisy data constrains the feasibility of training very large models with millions of parameters, as there is a risk of overfitting and requires extensive regularization or augmentation. In addition, the availability of standardized open source EEG datasets with diverse artifact types remains limited. This constrains the benchmarking and reproducibility of DL based denoising methods across various research groups. Another prominent challenge is interpretability, specifically understanding how the deep learning models make denoising decisions. Techniques such as attention maps and saliency visualization can be explored to reveal how models separate artifacts from neural activity.

Prominently, EEG denoising is not only about suppressing artifacts, but also requires the preservation of actual neural features. These features are crucial for downstream applications such as BCI control, clinical diagnosis, or cognitive state estimation. Hence, the model design must carefully balance artifact removal and signal integrity.

## 6 Future directions and emerging trends

With the growing adoption of EEG-based technologies in clinical diagnosis, brain computer interfaces, neurofeedback, and wearable neurotechnology, there is an increasing demand for robust, adaptable, and efficient EEG denoising methods. Future research must address the limitations of the current models while utilizing advancements in deep learning, signal representation, and computation. A promising direction lies in the development of self-supervised or unsupervised learning frameworks that can learn from

raw EEG data without requiring a clean ground truth. For instance, contrastive learning enables models to learn robust feature representations by distinguishing between similar and dissimilar EEG segments, while masked modelling and reconstruction-based losses can guide the network to focus on informative signal segments. Combining these techniques could improve the ability to model the nonlinear and nonstationary nature of EEG signals. Emerging solutions such as federated learning and neuromorphic computing may help mitigate some of the limitations. Federated learning enables collaborative model training across multiple institutions or subjects without requiring raw EEG data to be shared, thereby enhancing data privacy and inter-subject generalization. Meanwhile, neuromorphic computing powers brain inspired architectures to enable low latency and energy efficient EEG processing, making it promising for wearable denoising systems. Incorporating such techniques could improve model scalability, and real time applicability in future EEG denoising frameworks.

Recent advances in transformer architectures, spatiotemporal networks, and hybrid deep learning models have demonstrated improved performance in capturing complex temporal and spatial dependencies in EEG data. These architectures can model both local and global features across channels and time, offering enhanced artifact removal without distorting meaningful neural activity. Another crucial area of research is real-time denoising. For EEG systems to be feasible in mobile or embedded settings, denoising models must be computationally lightweight, support low processing delays, and maintain high signal quality. Hence, future EEG denoising lies in the design of intelligent, generalizable, and interpretable systems trained on diverse, real-world EEG data. These models must be integrated effortlessly into practical applications, enhancing the accuracy and reliability of EEG analysis across domains such as neurology, mental health, and human machine interaction.

## 7 Conclusion

EEG denoising is a critical pre-processing step in neurophysiological signal analysis, directly affecting the accuracy of subsequent clinical, cognitive, and BCI applications. Traditional signal processing methods, although widely used, often struggle to address the nonlinear and complex nature of real-world artifacts embedded within EEG recordings. This review presents a comprehensive overview of advanced deep learning (DL) models for EEG denoising, including CNNs, autoencoders, GANs, transformers, and hybrid models. Each architecture has unique strengths in terms of capturing spatiotemporal features and separating neural activity from noise. A dedicated performance analysis further evaluated these models under controlled settings using the EEGdenoiseNet dataset. Among them, CNN demonstrated superior performance for ocular artifacts, whereas the transformer excelled in denoising muscle related noise. Autoencoder offers a balance between accuracy and computational cost, and GAN, though flexible, requires greater tuning and longer training time.

This review also highlights key challenges in EEG denoising, such as the lack of ground truth, intersubject variability, and balancing artifact suppression with the preservation of neural information. These limitations underline the need for continued research on self-supervised or unsupervised learning techniques, domain adaptation, and lightweight architectures for real-time deployment. While this review discusses model generalizability and cross-subject adaptability extensively through an analysis of the existing

literature, these aspects were not explicitly verified through experimental validation in the present study. Future studies should aim to benchmark models for cross-subject and cross-dataset validations to establish robust generalizability in practical applications. In addition, this review aligns with the United Nations Sustainable Development Goal (SDG)–3 (Good Health and Well-being), by highlighting DL based EEG denoising frameworks that support improved neurological diagnostics, mental health monitoring, and brain computer interface technologies.

In conclusion, as EEG based technologies continue to expand into real-time and wearable domains, the need for robust, generalizable, and interpretable denoising models has become increasingly critical. Deep learning, with its capacity to model complex data representations and nonlinear patterns, is set to play a key role in enhancing the future of EEG artifact removal in both clinical and research environments.

## Supplementary Information

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Supplementary Material 1

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**Author contributions**
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**Data availability**
No datasets were generated or analysed during the current study.

## Declarations

**Ethical Statement**: This review article used the open source **EEGdenoiseNet dataset** to compare the performance of different deep learning models in denoising EEG signals, which is open-source and publicly available on GitHub (https://github.com/ncclabsustech/EEGdenoiseNet). No additional ethical approval was required for this review, as no new data involving human/animal subjects were collected.

**Consent to publish**
Not Applicable.

**Consent to participate**
Not Applicable.

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Not Applicable.

**Competing interest**
The authors declare no competing of interest.

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